

JEE MAIN REVISION MASTER: SEMICONDUCTOR SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 92 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	p-n Junction Diode & Zener Diode Regulation	46 PYQs	Semiconductor Diodes
Very High	Logic Gates & Truth Tables	46 PYQs	Digital Electronics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
p-n Junction Diode & Zener Diode Regulation	$\text{Efficiency}_{\text{half-wave}} = 40.6\%$, $\text{Efficiency}_{\text{full-wave}} = 81.2\%$, $V_L = V_Z = \text{const}$, $I_S = I_Z + I_L$
Logic Gates & Truth Tables	$\text{NAND: } Y = \overline{A \cdot B}$, $\text{NOR: } Y = \overline{A + B}$, $\overline{A \cdot B} = \overline{A} + \overline{B}$, $\overline{A + B} = \overline{A} \cdot \overline{B}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 p-n Junction Diode & Zener Diode Regulation (Very High Yield — 46 PYQs)

Core Principle: Forward and reverse characteristics of p-n junctions, rectifiers, ripple frequency, and Zener diode shunt voltage stabilization.

■■ **Caution / JEE Pitfall:** In full-wave rectifiers, output ripple frequency is TWICE the AC input frequency ($2f_{in}$), whereas for half-wave it is f_{in} .

■ **Speed Trick:** In reverse breakdown, Zener diode maintains constant voltage V_Z across its terminals as long as $I_Z \geq I_{Z,min}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : A diode under reverse-b..." [Ans: C]
- [2026 Morning Shift] Q#2 (MCQ): "For the given circuit (shown in part (A)) the time dependent input voltage $v_{in}(t)$ and corresponding output $v_o(t)$ are shown in part..." [Ans: B]
- [2026 Evening Shift] Q#2 (MCQ): "The charge stored by the capacitor C in the given circuit in the steady state is _____ μC ..." [Ans: C]

3.2 Logic Gates & Truth Tables (Very High Yield — 46 PYQs)

Core Principle: Universal logic gates (NAND and NOR), De Morgan's theorems, and Boolean circuit reduction.

■ **Caution / JEE Pitfall:** NAND and NOR gates are UNIVERSAL gates: any Boolean function can be constructed using only NAND or only NOR gates.

■ **Speed Trick:** Connect inputs of a NAND or NOR gate together to obtain a NOT gate: $\overline{A \cdot A} = \overline{A}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The output Y for the given inputs A and B to the circuit is :..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "For the given logic circuit, which of the following inputs combination will make both LED-1 and LED-2 to glow?..." [Ans: C]
- [2026 Evening Shift] Q#1 (MCQ): "If X and Y are the inputs, the given circuit works as _____." [Ans: D]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

For the given logic circuit, which of the following inputs combination will make both LED-1 and LED-2 to glow?

- (A) $A=0, B=1, C=1$
- (B) $A=1, B=0, C=0$
- (C) $A=1, B=0, C=1$ [Correct]
- (D) $A=1, B=1, C=0$

Correct Answer: C

Solution Analysis: Y_1 is the output of an OR gate whose inputs are A and B , $Y_1 = A + B$. Y_2 is an AND gate taking Y_1 and C as input $Y_2 = (A + B) \cdot C$. Y_3 is an AND gate taking Y_2 and A as input, $Y_3 = [(A + B) \cdot C] \cdot A$. $Y_3 = A \cdot (A + B) \cdot C = A \cdot C$. So, the truth table is,

A	B	C	Y_1	Y_2	Y_3
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	1	1

LED1 glows when $Y_1=1$ and LED2 glows when $Y_3=1$. So for both the LEDs to glow we need $Y_1=1$ and $Y_3=1$.

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Evening Shift (Q#2)

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R. Assertion A : A diode under reverse-biased condition provides very small current which is nearly independent of voltage until a critical limit at which the current increases drastically. Reason R : Below the critical voltage limit, only majority charge carriers flow which increases drastically above critical voltage. choose the correct answer from the options given below

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is NOT the correct explanation of A
- (C) is true but R is false [Correct]
- (D) is false but R is true

Correct Answer: C

Solution Analysis: In reverse bias, the positive terminal of the battery is connected to the n-side and the negative terminal to the p-side. This widens the depletion layer and increases the potential barrier. Only a very small current, known as the reverse saturation current, flows. This current is caused by the drift of minority charge carriers (electrons in p-type and holes in n-type) across the junction. Because the number of minority carriers is small and depends mainly on temperature, the current remains nearly constant (independent of voltage) until a high enough voltage, called the Zener or Breakdown voltage.